APARNA S

Palakkad, Kerala | 8714958606 | aparnas334455@gmail.com | linkedin.com/in/aparna-s-1ba391269

Master of Science in Microbiology

ABOUT ME

Research-driven M.Sc. Microbiologist with practical experience at premier institutions including BARC and ICMR-NIRT. Experienced in molecular biology, gamma radiation mutagenesis, microbial culturing, PAGE, and laboratory analytics. Dedicated to advancing medical microbiology and biotechnology innovation.

EDUCATION

Master of Science in Microbiology	Bharathidasan University, 8.98 CGPA Tiruchirapalli, Tamil Nadu	2024 – 2026
Bachelor of Science in Microbiology	Govt Arts & Science College 8.29 CGPA Kozhinjampara, Palakkad, University of Calicut, Kerala	2021 -2024

TECHNICAL SKILLS

- ✓ Microbial Isolation and Identification
- ✓ Microscopy
- ✓ Biochemical Assays and Analysis
- ✓ Antibiotic Susceptibility Testing
- ✓ DNA, RNA, and Plasmid Isolation
- ✓ Polymerase Chain Reaction (PCR)
- ✓ Gel Electrophoresis
- ✓ Protein Extraction
- ✓ Polyacrylamide Gel Electrophoresis (PAGE)
- ✓ Immunological Techniques
- ✓ Fermentation Studies
- ✓ Mutant Screening
- ✓ Zymography
- ✓ ELISA

SOFT SKILLS

- ✓ Aseptic Handling
- ✓ Scientific Documentation
- ✓ Report Writing
- ✓ Analytical Thinking
- ✓ Team Collaboration
- ✓ Problem-Solving Skills
- ✓ Experimental Planning
- ✓ MS Excel, Word & PowerPoint
- ✓ Adaptability and Quick Learning
- ✓ Laboratory Documentation
- ✓ Research Orientation
- ✓ Communication Skills

RESEARCH PROJECTS

MSc Dissertation Project

Title: Strain Improvement of *Saccharomyces cerevisiae* through Gamma Radiation for Bioethanol Production

- Performed gamma radiation-induced mutagenesis to develop ethanol-tolerant and thermotolerant yeast strains for enhanced bioethanol production.
- Screened irradiated strains under varying ethanol concentrations to evaluate improved ethanol tolerance and stress adaptability.
- Assessed fermentation efficiency and ethanol production of mutant strains under different glucose concentrations and temperature conditions.
- Conducted comparative analysis of wild-type and mutant yeast strains using microbial growth studies and enzyme assays.

BSc Project

Title: Investigating the Drug-Resistant Pattern of Vancomycin-Resistant Bacteria Isolated from Various Sources of the Palakkad Region

- Isolated bacterial strains from different water sources in the Palakkad region using standard microbiological techniques.
- Evaluated the antimicrobial resistance patterns of bacterial isolates through antibiotic susceptibility testing.
- Performed cultural and biochemical characterisation of antibiotic-resistant bacterial isolates for preliminary identification.
- Conducted molecular-level identification of the most resistant bacterial strain using molecular biology techniques.

INTERNSHIPS

Bhabha Atomic Research Centre | Mumbai | 01 Jan 2026 – 15 Apr 2026
Food Corporation of India | Chennai | 04 Jun 2025 – 13 Jun 2025
ICMR-National Institute for Research in Tuberculosis, Department of Virology | Chennai13 | May 2025 – 02 Jun 2025

ACTIVITIES & CERTIFICATIONS

Annual Member of **Microbiologists Society**, India (MBSI).
Attended **National Seminar** on Sustainable Future through Microbial Innovations.
Attended **National Seminar** on Human, Animal, Plant-Vector Borne Diseases and their Control.
Attended **International Experts'** Talk Show on Bioprocessing of Secondary Metabolites: Current Challenges, Emerging Opportunities, and Future Outlook.
Participated in a **Workshop** on Chlorella genomics, cultivation, isolation, and value-added products.
Completed **NPTEL** course on **RNA Biology** (Elite) & **Research Methodology** (Elite)
Attended Science Academies' Lecture Workshop on **Recent Advances in Biomedical Sciences for Accelerating Drug Discovery and Therapeutics**, 20–21 July 2026

AREA OF INTERESTS

- Clinical and Medical Microbiology
- Molecular Biology and Microbial Genetics
- Microbiome Studies
- Virology and Emerging Infectious Diseases
- Immunology and Host–Pathogen Interactions
- Bioinformatics and Microbial Genomics

LANGUAGES

English
Malayalam
Tamil

REFERENCES

Dr. D. Dhanasekaran
Head of Department, Department of Microbiology,
Bharathidasan University-Trichy
Tamilnadu

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